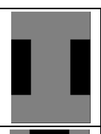
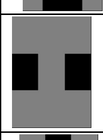
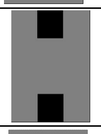
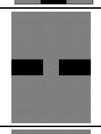
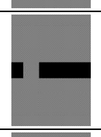
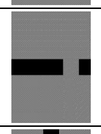
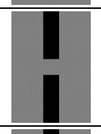
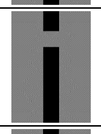
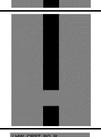
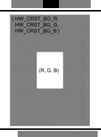
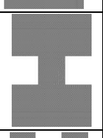
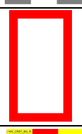
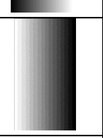
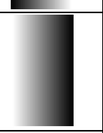
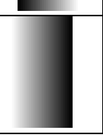
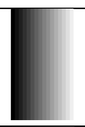
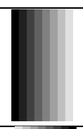
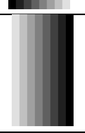
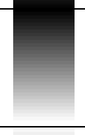
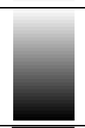
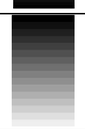
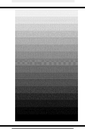
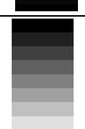
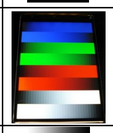
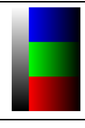
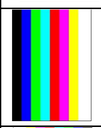
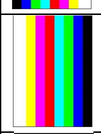
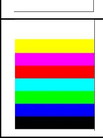
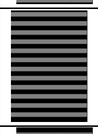
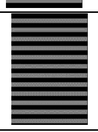
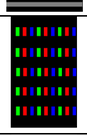
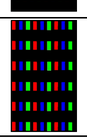
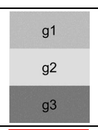
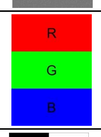
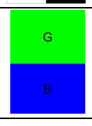
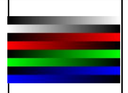
Total 156

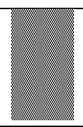
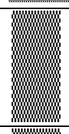
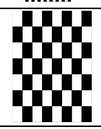
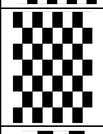
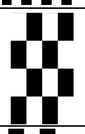
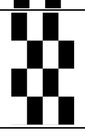
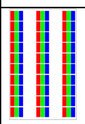
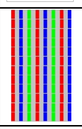
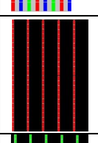
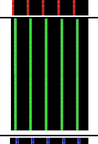
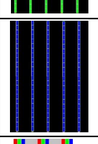
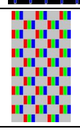
1、显示pattern的命令格式为：FpgaDisplay(ptnNum, R, G, B)

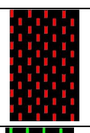
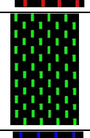
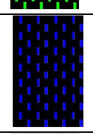
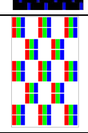
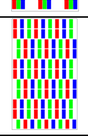
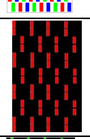
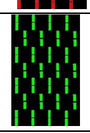
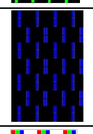
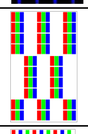
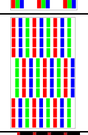
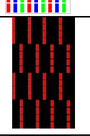
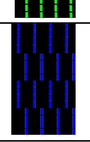
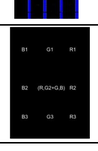
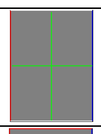
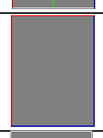
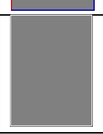
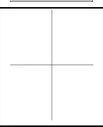
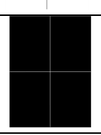
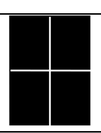
	pattern num	pattern num(HEX)	pattern description	pattern view	command description	Remark	串口代码
	picNum		图片		FpgaPic(picNum)	ptnNum从0开始计数	第1张图：AA 10 00 00 0F 00 12 48 第2张图：AA 10 00 01 0F 00 12 48 第3张图：AA 10 00 02 0F 00 12 48
纯色							
	0	0	纯色画面		FpgaDisplay(0, R, G, B)	0表示纯色画面 R/G/B为对应的颜色值	AA 02 00 00 00 R 00 G 00 B 0F 00 12 48
	0	0	red		FpgaDisplay(0, 255, 0, 0)		AA 02 00 00 00 FF 00 00 00 00 0F 00 12 48
	0	0	green		FpgaDisplay(0, 0, 255 ,0)		AA 02 00 00 00 00 00 FF 00 00 0F 00 12 48
	0	0	blue		FpgaDisplay(0, 0, 0, 255)		AA 02 00 00 00 00 00 00 00 FF 0F 00 12 48
	0	0	blank		FpgaDisplay(0, 0, 0, 0)		AA 02 00 00 00 0000 00 00 00 0F 00 12 48
	0	0	white		FpgaDisplay(0, 255, 255, 255)		AA 02 00 00 00 FF 00 FF 00 FF 0F 00 12 48
	0	0	gray128		FpgaDisplay(0, 128, 128, 128)	灰画面灰阶请与crosstalk底色保持一致	AA 02 00 00 00 80 00 80 00 80 0F 00 12 48
crosstalk							
	1	1	crosstalk, huawei spec, middle area black		FpgaDisplay(1, val, xx, xx)	中心窗为黑色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/3v, 2/3v	4 1/3h 2/3h 1/3v 2/3v 0 rval AA 02 00 01 00 7F 0F 00 12 48
	7	7	crosstalk, huawei spec, middle area white		FpgaDisplay(7, val, xx, xx)	中心窗为白色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/3v, 2/3v	4 1/3h 2/3h 1/3v 2/3v 255 rval AA 02 00 07 00 7F 0F 00 12 48
	2	2	crosstalk, samsung spec, up & down area black		FpgaDisplay(2, val, xx, xx)	上下窗为黑色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/4v, 3/4v	3 1/3h 2/3h 1/4v 3/4v 0 rval AA 02 00 02 00 7F 0F 00 12 48
	8	8	crosstalk, samsung spec, up & down area white		FpgaDisplay(8, val, xx, xx)	上下窗为白色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/4v, 3/4v	3 1/3h 2/3h 1/4v 3/4v 255 rval AA 02 00 08 00 7F 0F 00 12 48
	4	4	crosstalk, samsung spec, left & right area black		FpgaDisplay(4, val, xx, xx)	左右窗为黑色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/3v, 2/3v	2 1/3h 2/3h 1/3v 2/3v 0 rval AA 02 00 04 00 7F 0F 00 12 48
	10	A	crosstalk, samsung spec, left & right area white		FpgaDisplay(10, val, xx, xx)	左右窗为白色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/3v, 2/3v	2 1/3h 2/3h 1/3v 2/3v 255 rval AA 02 00 0A 00 7F 0F 00 12 48
	3	3	crosstalk, moto spec, up & down area black		FpgaDisplay(3, val, xx, xx)	上下窗为黑色，其余部分灰阶为val 水平分界：1/6h, 5/6h；竖直分界：1/4v, 3/4v	3 1/6h 5/6h 1/4v 3/4v 0 rval AA 02 00 03 00 7F 0F 00 12 48
	9	9	crosstalk, moto spec, up & down area white		FpgaDisplay(9, val, xx, xx)	上下窗为白色，其余部分灰阶为val 水平分界：1/6h, 5/6h；竖直分界：1/4v, 3/4v	3 1/6h 5/6h 1/4v 3/4v 255 rval AA 02 00 09 00 7F 0F 00 12 48
	5	5	crosstalk, moto spec, left & right area black		FpgaDisplay(5, val, xx, xx)	左右窗为黑色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/4v, 3/4v	2 1/3h 2/3h 1/4v 3/4v 0 rval AA 02 00 05 00 7F 0F 00 12 48
	11	B	crosstalk, moto spec, left & right area white		FpgaDisplay(11, val, xx, xx)	左右窗为白色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/4v, 3/4v	2 1/3h 2/3h 1/4v 3/4v 255 rval AA 02 00 0B 00 7F 0F 00 12 48
	6	6	crosstalk, moto spec, four corner area black		FpgaDisplay(6, val, xx, xx)	四角窗为黑色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/4v, 3/4v	1 1/3h 2/3h 1/4v 3/4v 0 rval AA 02 00 06 00 7F 0F 00 12 48
	12	C	crosstalk, moto spec, four corner area white		FpgaDisplay(12, val, xx, xx)	四角窗为白色，其余部分灰阶为val 水平分界：1/3h, 2/3h；竖直分界：1/4v, 3/4v	1 1/3h 2/3h 1/4v 3/4v 255 rval AA 02 00 0C 00 7F 0F 00 12 48
	94	5E	crosstalk, asus spec, middle area		FpgaDisplay(94, val1, val2, xx)	中心窗灰阶为val2，其余部分灰阶为val1 水平分界：1/4h,3/4h；竖直分界：1/4v,3/4v	4 1/4h 3/4h 1/4v 3/4v gval rval AA 02 00 5E 00 7F 00 00 0F 00 12 48

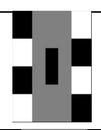
95	5F	crosstalk, asus spec, left & right area		FpgaDisplay(95, val1, val2, xx)	左右窗灰阶为val2，其余部分灰阶为val1 水平分界：1/4h,3/4h；竖直分界：1/4v,3/4v	2	1/4h	3/4h	1/4v	3/4v	gval	rval	AA 02 00 5F 00 7F 00 00 0F 00 12 48
96	60	crosstalk, asus spec, up & down area		FpgaDisplay(96, val1, val2, xx)	上下窗灰阶为val2，其余部分灰阶为val1 水平分界：1/4h,3/4h；竖直分界：1/4v,3/4v	3	1/4h	3/4h	1/4v	3/4v	gval	rval	AA 02 00 60 00 7F 00 00 0F 00 12 48
107	6B	crosstalk, samsung spec, left & right area		FpgaDisplay(107, val1,val2,val3)	h: 1/3 v: 1/3 val1:设置底色灰阶，三星规格128 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定框垂直方向起点	2	1/3h	2/3h	{gval[3:0,bval[7:0]}	y1+v/3	gval[7]	rval	AA 02 00 6B 00 80 00 00 00 50 0F 00 12 48
108	6C	crosstalk, samsung spec, up & down area		FpgaDisplay(108, val1,val2, val3)	h: 1/3 v: 1/3 val1:设置底色灰阶，三星规格128 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定框水平方向起点	3	{gval[3:0,bval[7:0]}	x1+1/3h	1/3v	2/3v	gval[7]	rval	AA 02 00 6C 00 80 00 00 00 80 0F 00 12 48
117	75	crosstalk，meizu left & right		FpgaDisplay(117, val1,val2,val3)	val1:设置底色灰阶，魅族规格127 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定测量方块的边长的一半	2	1/2h-{gval[3:0,bval[7:0]}	1/2h+{gval[3:0,bval[7:0]}	1/2v-{gval[3:0,bval[7:0]}	1/2v+{gval[3:0,bval[7:0]}	gval[7]	rval	AA 02 00 75 00 7F 00 00 00 20 0F 00 12 48
118	76	crosstalk，meizu left & right		FpgaDisplay(118, val1,val2,val3)	val1:设置底色灰阶，魅族规格127 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定测量方块的边长的一半	2	1/4h-{gval[3:0,bval[7:0]}	1/4h+{gval[3:0,bval[7:0]}	1/2v-{gval[3:0,bval[7:0]}	1/2v+{gval[3:0,bval[7:0]}	gval[7]	rval	AA 02 00 76 00 7F 00 00 00 20 0F 00 12 48
119	77	crosstalk，meizu left & right		FpgaDisplay(119, val1,val2,val3)	val1:设置底色灰阶，魅族规格127 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定测量方块的边长的一半	2	3/4h-{gval[3:0,bval[7:0]}	3/4h+{gval[3:0,bval[7:0]}	1/2v-{gval[3:0,bval[7:0]}	1/2v+{gval[3:0,bval[7:0]}	gval[7]	rval	AA 02 00 77 00 7F 00 00 00 20 0F 00 12 48
120	78	crosstalk，meizu up & down		FpgaDisplay(120, val1,val2,val3)	val1:设置底色灰阶，魅族规格127 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定测量方块的边长的一半	3	1/2h-{gval[3:0,bval[7:0]}	1/2h+{gval[3:0,bval[7:0]}	1/2v-{gval[3:0,bval[7:0]}	1/2v+{gval[3:0,bval[7:0]}	gval[7]	rval	AA 02 00 78 00 7F 00 00 00 20 0F 00 12 48
121	79	crosstalk，meizu up & down		FpgaDisplay(121, val1,val2,val3)	val1:设置底色灰阶，魅族规格127 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定测量方块的边长的一半	3	1/2h-{gval[3:0,bval[7:0]}	1/2h+{gval[3:0,bval[7:0]}	1/4v-{gval[3:0,bval[7:0]}	1/4v+{gval[3:0,bval[7:0]}	gval[7]	rval	AA 02 00 79 00 7F 00 00 00 20 0F 00 12 48
122	7A	crosstalk，meizu up & down		FpgaDisplay(122, val1,val2,val3)	val1:设置底色灰阶，魅族规格127 val2[7]:设置框颜色，0=黑，1=白 {val2[3:0],val3}:设定测量方块的边长的一半	3	1/2h-{gval[3:0,bval[7:0]}	1/2h+{gval[3:0,bval[7:0]}	3/4v-{gval[3:0,bval[7:0]}	3/4v+{gval[3:0,bval[7:0]}	gval[7]	rval	AA 02 00 7A 00 7F 00 00 00 20 0F 00 12 48
128	80	华为crosstalk 新规格		FpgaDisplay(128, R, G, B)	请在ARM程序的“include.h”文件中设置底色 (HW_CRST_BG_R，HW_CRST_BG_G，HW_CRST_BG_B)	4	1/3h	2/3h	1/3v	2/3v	{rval, gval, bval}	{bg_r,bg_g,bg_b}	AA 02 00 80 00 FF 00 00 00 00 0F 00 12 48
134	86	crosstalk, sony spec, left & right area black		FpgaDisplay(134, val1, val2, xx)	h: 1/3 v: 2/8 矩形框灰阶val2可调 底色灰阶val1可调，厂内规格127，如客户无特别要求，请使用127	2	1/3h	2/3h	3/8v	5/8v	gval	rval	AA 02 00 86 00 7F 00 FF 0F 00 12 48
135	87	crosstalk, sony spec, up & down area black		FpgaDisplay(135, val1, val2, xx)	h: 1/3 v: 3/8 矩形框灰阶val2可调 底色灰阶val1可调，厂内规格127，如客户无特别要求，请使用127	3	1/3h	2/3h	3/8v	5/8v	gval	rval	AA 02 00 87 00 7F 00 FF 0F 00 12 48
132	84	华为红框		FpgaDisplay(132, val1,val2,val3)	val1:设置中心底色灰阶，厂内规格255 {val2[1:0],val3}:设定红框宽度	4	{gval[1:0],bval[7:0]}	h-x1	{gval[1:0],bval[7:0]}	v-y1	rval	{255,0,0}	AA 02 00 84 00 FF 00 00 00 50 0F 00 12 48
133	85	华为红十字		FpgaDisplay(133, R, G, B)	请在ARM程序的“include.h”文件中设置底色 (HW_CROSS_BG_R，HW_CROSS_BG_G，HW_CROSS_BG_B)	1	1/3h	2/3h	1/3v	2/3v	{bg_r,bg_g,bg_b}	{rval, gval, b_val}	AA 02 00 85 00 FF 00 00 00 00 0F 00 12 48
bar(条块)															
	17	11	v256-graybar （黑-白）		FpgaDisplay(17, xx, xx, xx)								AA 02 00 11 0F 00 12 48		
	35	23	v256-graybar （白-黑）		FpgaDisplay(35, xx, xx, xx)								AA 02 00 23 0F 00 12 48		
	45	2D	v64-graybar （黑-白）		FpgaDisplay(45, xx, xx, xx)								AA 02 00 2D 0F 00 12 48		
	47	2F	v64-graybar （白-黑）		FpgaDisplay(47, xx, xx, xx)								AA 02 00 2F 0F 00 12 48		
	146	92	纵向64灰阶过渡		FpgaDisplay(146, val1, xx, xx)	纵向64灰阶过渡，起始灰阶val1，step为1							AA 02 00 92 00 00 0F 00 12 48		
	41	29	v32-graybar （黑-白）		FpgaDisplay(41, xx, xx, xx)								AA 02 00 29 0F 00 12 48		
	43	2B	v32-graybar （白-黑）		FpgaDisplay(43, xx, xx, xx)								AA 02 00 2B 0F 00 12 48		

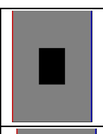
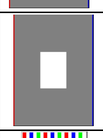
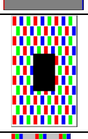
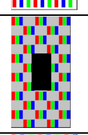
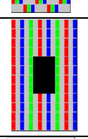
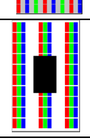
147	93	纵向32灰阶过渡		FpgaDisplay(147, val1, xx, xx)	纵向32灰阶过渡，起始灰阶val1，step为1	AA 02 00 93 00 00 0F 00 12 48
15	F	v16-graybar （黑-白）		FpgaDisplay(15, xx, xx, xx)		AA 02 00 0F 0F 00 12 48
33	21	v16-graybar （白-黑）		FpgaDisplay(33, xx, xx, xx)		AA 02 00 21 0F 00 12 48
148	94	纵向16灰阶过渡		FpgaDisplay(148, val1, xx, xx)	纵向16灰阶过渡，起始灰阶val1，step为1	AA 02 00 94 00 00 0F 00 12 48
37	25	v8-graybar （黑-白）		FpgaDisplay(37, xx, xx, xx)		AA 02 00 25 0F 00 12 48
39	27	v8-graybar （白-黑）		FpgaDisplay(39, xx, xx, xx)		AA 02 00 27 0F 00 12 48
149	95	纵向8灰阶过渡		FpgaDisplay(149, val1, xx, xx)	纵向8灰阶过渡，起始灰阶val1，step为1	AA 02 00 95 00 00 0F 00 12 48
150	96	纵向4灰阶过渡		FpgaDisplay(150, val1, xx, xx)	纵向4灰阶过渡，起始灰阶val1，step为1	AA 02 00 96 00 00 0F 00 12 48
151	97	纵向2灰阶过渡		FpgaDisplay(151, val1, xx, xx)	纵向2灰阶过渡，起始灰阶val1，step为1	AA 02 00 97 00 00 0F 00 12 48
18	12	h256-graybar （黑-白）		FpgaDisplay(18, xx, xx, xx)		AA 02 00 12 0F 00 12 48
36	24	h256-graybar （白-黑）		FpgaDisplay(36, xx, xx, xx)		AA 02 00 24 0F 00 12 48
46	2E	h64-graybar （黑-白）		FpgaDisplay(46, xx, xx, xx)		AA 02 00 2E 0F 00 12 48
48	30	h64-graybar （白-黑）		FpgaDisplay(48, xx, xx, xx)		AA 02 00 30 0F 00 12 48
42	2A	h32-graybar （黑-白）		FpgaDisplay(42, xx, xx, xx)		AA 02 00 2A 0F 00 12 48
44	2C	h32-graybar （白-黑）		FpgaDisplay(44, xx, xx, xx)		AA 02 00 2C 0F 00 12 48
16	10	h16-graybar （黑-白）		FpgaDisplay(16, xx, xx, xx)		AA 02 00 10 0F 00 12 48
34	22	h16-graybar （白-黑）		FpgaDisplay(34, xx, xx, xx)		AA 02 00 22 0F 00 12 48
38	26	h8-graybar （黑-白）		FpgaDisplay(38, xx, xx, xx)		AA 02 00 26 0F 00 12 48
40	28	h8-graybar （白-黑）		FpgaDisplay(40, xx, xx, xx)		AA 02 00 28 0F 00 12 48
72	48	colorbar （HTC SPEC）		FpgaDisplay(72, xx, xx, xx)		AA 02 00 48 0F 00 12 48
92	5C	colorbar （ASUS SPEC）		FpgaDisplay(92, xx, xx, xx)	256阶	AA 02 00 5C 0F 00 12 48
106	6A	colorbar （Lenovo SPEC）		FpgaDisplay(106, xx, xx, xx)	64阶	AA 02 00 6A 0F 00 12 48
125	7D	gray RGB bar		FpgaDisplay(125, xx, xx, xx)	256阶	AA 02 00 7D 0F 00 12 48
138	8A	hw colour bar gradients		FpgaDisplay(138, xx, xx, xx)	华为彩阶渐变，256阶	AA 02 00 8A 0F 00 12 48
29	1D	灰阶+RGB渐变		FpgaDisplay(29, xx, xx, xx)	32灰阶，R/G/B 32阶渐变	AA 02 00 1D 0F 00 12 48

	90	5A	灰阶+RGB渐变		FpgaDisplay(90, xx, xx, xx)	256灰阶，R/G/B 256阶渐变	AA 02 00 5A 0F 00 12 48
	13	D	v-coloubar （黑-白）		FpgaDisplay(13, xx, xx, xx)		AA 02 00 0D 0F 00 12 48
	49	31	v-coloubar （白-黑）		FpgaDisplay(49, xx, xx, xx)		AA 02 00 31 0F 00 12 48
	14	E	h-coloubar （黑-白）		FpgaDisplay(14, xx, xx, xx)		AA 02 00 0E 0F 00 12 48
	50	32	h-coloubar （白-黑）		FpgaDisplay(50, xx, xx, xx)		AA 02 00 32 0F 00 12 48
	32	20	1x1bar （黑灰黑）		FpgaDisplay(32, val, xx, xx)	奇数行固定为黑色，偶数行为val。 val表示灰度值，灰度值可调	AA 02 00 20 00 7F 0F 00 12 48
	69	45	1X1bar		FpgaDisplay(69, val1, val2, xx)	奇数行为灰阶为val1，偶数行灰阶为val2	AA 02 00 45 00 FF 00 00 0F 00 12 48
	104	68	1x1bar		FpgaDisplay(104, R, G, B)	奇数行为黑色，偶数行为{R,G,B}	AA 02 00 68 00 7F 00 7F 00 7F 0F 00 12 48
	105	69	1X1bar		FpgaDisplay(105, R, G, B)	奇数行为{R,G,B}，偶数行为黑色	AA 02 00 69 00 7F 00 7F 00 7F 0F 00 12 48
	75	4B	1x1 dot bar		FpgaDisplay(75，val, xx, xx)	奇数行：黑色 偶数行：交替为{0,val,0},{val,0,val}	AA 02 00 4B 00 7F 0F 00 12 48
	76	4C	1x1 dot bar		FpgaDisplay(76，val, xx, xx)	奇数行：交替为{val,0,val},{0,val,0} 偶数行：黑色	AA 02 00 4C 00 7F 0F 00 12 48
	144	90	1x1 dot bar		FpgaDisplay(144, val1, val2, 0)	奇数行：{val1, val2, val1} 偶数行：{val2, val1, val2}	AA 02 00 90 00 FF 00 00 0F 00 12 48
	126	7E	垂直方向3等分灰阶可定义bar		FpgaDisplay(126, g1, g2, g3)		AA 02 00 7E 00 80 00 A0 00 60 0F 00 12 48
	127	7F	RGB水平bar，RGB亮度可定义		FpgaDisplay(127, R, G, B)		AA 02 00 7F 00 FF 00 FF 00 FF 0F 00 12 48
	130	82	左黑右白		FpgaDisplay(130, xx, xx, xx)		AA 02 00 82 0F 00 12 48
	131	83	左白右黑		FpgaDisplay(131, xx, xx, xx)		AA 02 00 83 0F 00 12 48
	142	8E	转厂武汉半绿半蓝		FpgaDisplay(142, xx, G, B)		AA 02 00 8E 00 00 00 FF 00 FF 0F 00 12 48
	146	92	竖直彩阶bar（上海）		FpgaDisplay(146, xx, xx, xx)		AA 02 00 92 0F 00 12 48
	147	93	水平渐变bar		FpgaDisplay(147, xx, xx, xx)		AA 02 00 93 0F 00 12 48
棋盘格							
	19	13	chessboard （64*64）		FpgaDisplay(19, xx, xx, xx)	左上角为白色	AA 02 00 13 0F 00 12 48

	52	34	chessboard （64*64）		FpgaDisplay(52, xx, xx, xx)	左上角为黑色	AA 02 00 34 0F 00 12 48
	55	37	chessboard （32*32）		FpgaDisplay(55, xx, xx, xx)	左上角为白色	AA 02 00 37 0F 00 12 48
	56	38	chessboard （32*32）		FpgaDisplay(56, xx, xx, xx)	左上角为黑色	AA 02 00 38 0F 00 12 48
	53	35	chessboard （16*16）		FpgaDisplay(53, xx, xx, xx)	左上角为白色	AA 02 00 35 0F 00 12 48
	54	36	chessboard （16*16）		FpgaDisplay(54, xx, xx, xx)	左上角为黑色	AA 02 00 36 0F 00 12 48
	20	14	chessboard （8*8）		FpgaDisplay(20, xx, xx, xx)	左上角为白色	AA 02 00 14 0F 00 12 48
	51	33	chessboard （8*8）		FpgaDisplay(51, xx, xx, xx)	左上角为黑色	AA 02 00 33 0F 00 12 48
	70	46	chessboard （4*4）		FpgaDisplay(70, xx, xx, xx)	左上角为白色	AA 02 00 46 0F 00 12 48
	71	47	chessboard （4*4）		FpgaDisplay(71, xx, xx, xx)	左上角为黑色	AA 02 00 47 0F 00 12 48
	77	4D	chessboard （8*6）		FpgaDisplay(77, xx, xx, xx)	左上角为白色	AA 02 00 4D 0F 00 12 48
	78	4E	chessboard （8*6）		FpgaDisplay(78, xx, xx, xx)	左上角为黑色	AA 02 00 4E 0F 00 12 48
	93	5D	chessboard （20*12）		FpgaDisplay(93, val, xx, xx)	val表示左上角第一个格子的灰度 0：黑色 255：白色	AA 02 00 5D 00 FF 0F 00 12 48
	111	6F	chessboard		FpgaDisplay(111,val1,val2, val3)	左上角第一个格子的灰度：val1 左上角第二个格子的灰度：255-val1 val2 = 水平方向每个格子所占像素数 val3 = 垂直方向每个格子所占像素数	AA 02 00 6F 00 FF 00 80 00 80 0F 00 12 48
点类(checker)							
	21	15	flicker - pixcel column inversion		FpgaDisplay(21, val, xx, xx)	奇数列：灰阶为val 偶数列：黑色	AA 02 00 15 00 7F 0F 00 12 48
	22	16	flicker -column inversion		FpgaDisplay(22, val, xx, xx)	奇数列：{val, 0, val} 偶数列：{0, val, 0}	AA 02 00 16 00 7F 0F 00 12 48
	60	3C	flicker - R column inversion		FpgaDisplay(60, val, xx, xx)	奇数列：{val, 0, 0l} 偶数列：{0, 0, 0}	AA 02 00 3C 00 7F 0F 00 12 48
	64	40	flicker - G column inversion		FpgaDisplay(64, val, xx, xx)	奇数列：{0, val, 0l} 偶数列：{0, 0, 0}	AA 02 00 40 00 7F 0F 00 12 48
	68	44	flicker - B column inversion		FpgaDisplay(68, val, xx, xx)	奇数列：{0, 0, val} 偶数列：{0, 0, 0}	AA 02 00 44 00 7F 0F 00 12 48
	23	17	flicker - 1 pixel inversion		FpgaDisplay(23, val, xx, xx)	以pixel为单位翻转： {val, val, val}和{0, 0, 0}	AA 02 00 17 00 7F 0F 00 12 48
	140	8C	RGBW-JDI white pixel白画面		FpgaDisplay(140, R, G, B)	以pixel为单位翻转： {0, 0, 0}和{R, G, B}	AA 02 00 8C 00 7F 00 7F 00 7F 0F 00 12 48
	141	8D	RGBW-JDI RGB pixel白画面		FpgaDisplay(141, R, G, B)	以pixel为单位翻转： {R, G, B}和{R, G, 0}	AA 02 00 8D 00 7F 00 7F 00 7F 0F 00 12 48
	24	18	flicker - 1dot inversion		FpgaDisplay(24, val, xx, xx)	以pixel为单位翻转： {val, 0, val}和{0, val, 0}	AA 02 00 18 00 7F 0F 00 12 48

	57	39	flicker - R 1 dot inversion		FpgaDisplay(57, val, xx, xx)	以pixel为单位翻转： {val, 0, 0}和{0, 0, 0}	AA 02 00 39 00 7F 0F 00 12 48
	61	3D	flicker - G 1 dot inversion		FpgaDisplay(61, val, xx, xx)	以pixel为单位翻转： {0, val, 0}和{0, 0, 0}	AA 02 00 3D 00 7F 0F 00 12 48
	65	41	flicker - B 1 dot inversion		FpgaDisplay(65, val, xx, xx)	以pixel为单位翻转： {0, 0, val}和{0, 0, 0}	AA 02 00 41 00 7F 0F 00 12 48
	25	19	flicker - 2 pixel inversion		FpgaDisplay(25, val, xx, xx)	以1x2 pixel为单位翻转： {val, val, val}和{0, 0, 0}	AA 02 00 19 00 7F 0F 00 12 48
	26	1A	flicker - 2 dot inversion		FpgaDisplay(26, val, xx, xx)	以1x2 pixel为单位翻转： {val, 0, val}和{0, val, 0}	AA 02 00 1A 00 7F 0F 00 12 48
	58	3A	flicker - R 2 dot inversion		FpgaDisplay(58, val, xx, xx)	以1x2 pixel为单位翻转： {val, 0, 0}和{0, 0, 0}	AA 02 00 3A 00 7F 0F 00 12 48
	62	3E	flicker - G 2 dot inversion		FpgaDisplay(62, val, xx, xx)	以1x2 pixel为单位翻转： {0, val, 0}和{0, 0, 0}	AA 02 00 3E 00 7F 0F 00 12 48
	66	42	flicker - R 2 dot inversion		FpgaDisplay(66, val, xx, xx)	以1x2 pixel为单位翻转： {0, 0, val}和{0, 0, 0}	AA 02 00 42 00 7F 0F 00 12 48
	27	1B	flicker - 4 pixel inversion		FpgaDisplay(27, val, xx, xx)	以1x4 pixel为单位翻转： {val, val, val}和{0, 0, 0}	AA 02 00 1B 00 7F 0F 00 12 48
	28	1C	flicker - 4 dot inversion		FpgaDisplay(28, val, xx, xx)	以1x4 pixel为单位翻转： {val, 0, val}和{0, val, 0}	AA 02 00 1C 00 7F 0F 00 12 48
	59	3B	flicker - R 4 dot inversion		FpgaDisplay(59, val, xx, xx)	以1x4 pixel为单位翻转： {val, 0, 0}和{0, 0, 0}	AA 02 00 3B 00 7F 0F 00 12 48
	63	3F	flicker - G 4 dot inversion		FpgaDisplay(63, val, xx, xx)	以1x4 pixel为单位翻转： {0, val, 0}和{0, 0, 0}	AA 02 00 3F 00 7F 0F 00 12 48
	67	43	flicker - B 4 dot inversion		FpgaDisplay(67, val, xx, xx)	以1x4 pixel为单位翻转： {0, 0, val}和{0, 0, 0}	AA 02 00 43 00 7F 0F 00 12 48
	82	52	Bright dot test		FpgaDisplay(82, R, G, B)	第1排：{0,0,dot_b1},{0,dot_g1,0},{dot_r1,0,0} 第2排：{0,0,dot_b2},{R,dot_g2+R,B},{dot_r2,0,0} 第3排：{0,0,dot_b3},{0,dot_g3,0},{dot_r3,0,0} 注： dot_r1,dot_g1,dot_b1,dot_r2,dot_g2,dot_b2,dot_r3,dot_g3,dot_b3为ARM程序中宏定义	AA 02 00 52 00 00 00 00 00 0F 00 12 48
	143	8F	转厂武汉白底暗点		FpgaDisplay(143, R, G, B)	屏幕中心颜色{R, G, B}，其余为白色	AA 02 00 8F 00 00 00 00 00 00 0F 00 12 48
线类							
	31	1F	边框检测		FpgaDisplay(31, val, xx, xx)	左红右蓝上下白绿十字	AA 02 00 1F 00 7F 0F 00 12 48
	80	50	蓝红框黑底画面		FpgaDisplay(80, val, xx, xx)	上红左红下蓝右蓝 其余部分由val指定	AA 02 00 50 00 7F 0F 00 12 48
	81	51	边框检测		FpgaDisplay(81, val, val1, val2)	val表示底色灰度值，灰度值可调 val1[6:4]定义上边线颜色，如100=red val1[2:0]定义下边线颜色,如010=green val2[6:4]定义左边线颜色,如001=blue val2[2:0]定义右边线颜色,如110=yellow	AA 02 00 51 00 7F 00 77 00 41 0F 00 12 48
	88	58	cross at center		FpgaDisplay(88, val, xx, xx)	黑色十字，底色val可调	AA 02 00 58 00 FF 0F 00 12 48
	89	59	cross at center		FpgaDisplay(89, val, xx, xx)	白色十字，底色val可调	AA 02 00 59 00 00 0F 00 12 48
	109	6D	AOI镜头校准画面		FpgaDisplay(109, xx, xx, xx)	周边10行白，中间为棋盘格（HD:13*7，FHD:15*9）	暂未实现	AA 02 00 6D 00 00 00 00 00 00 0F 00 12 48 或者 AA 02 00 6D 0F 00 12 48
	115	73	边框检测组装精度 （sony 规格）		FpgaDisplay(115, val, val1, val2)	val表示底色灰度值，灰度值可调 val1[6:4]定义上边线颜色，如100=red val1[2:0]定义下边线颜色,如010=green val2[6:4]定义左边线颜色,如001=blue val2[2:0]定义右边线颜色,如110=yellow 中间2行2列为白色	AA 02 00 73 00 00 00 77 00 41 0F 00 12 48

	129	81	中心红框对位画面		FpgaDisplay(129, val1,val2,val3)	val1:设置底色灰阶，厂内规格0 {val2[1:0],val3}:设定红框的边长		AA 02 00 81 00 00 00 00 00 80 0F 00 12 48
	145	91	对位画面		FpgaDisplay(145, val1,val2,xx)	val1: 设置线条颜色 val2: 设置背景色		AA 02 00 91 00 FF 00 00 00 80 0F 00 12 48
字符显示								
	74	4A	Full screen character （HTC SPEC）		FpgaDisplay(74，val, xx, xx)	字符为白色，背景色为val		AA 02 00 4A 00 7F 0F 00 12 48
	110	6E	Full screen character （HTC SPEC）		FpgaDisplay(110, val1,val2, val3)	字符为白色，背景色为val val2表示水平方向放大倍数 val3表示垂直方向放大倍数	放大倍数为1有问题	AA 02 00 6E 00 80 00 02 00 02 0F 00 12 48
	84	54	No OTP pattern		FpgaDisplay(84, val, xx, xx)	字符为红色，背景色为val		AA 02 00 54 00 00 0F 00 12 48
	116	74	崩角报警画面		FpgaDisplay(116, val, xx, xx)	字符为红色，背景色为val		AA 02 00 74 00 00 0F 00 12 48
	139	8B	“开盖”		FpgaDisplay(139, val, xx, xx)	字符为红色，背景色为val		AA 02 00 8B 00 00 0F 00 12 48
	85	55	NG 显示在上半屏		FpgaDisplay(85, val, xx, xx)	字符为红色，背景色为val		AA 02 00 55 00 00 0F 00 12 48
	114	72	NG 显示在中间		FpgaDisplay(114, val, xx, xx)	字符为红色，背景色为val		AA 02 00 72 00 00 0F 00 12 48
	113	71	NG 显示在下半屏		FpgaDisplay(113, val, xx, xx)	字符为红色，背景色为val		AA 02 00 71 00 00 0F 00 12 48
	86	56	ID ERROR		FpgaDisplay(86, val, xx, xx)	字符为红色，背景色为val		AA 02 00 56 00 00 0F 00 12 48
	87	57	ID RESET		FpgaDisplay(87, val, xx, xx)	字符为红色，背景色为val		AA 02 00 57 00 00 0F 00 12 48
	97	61	VSN VSP		FpgaDisplay(97, val, xx, xx)	字符为红色，背景色为val		AA 02 00 61 00 00 0F 00 12 48
	98	62	VSN		FpgaDisplay(98, val, xx, xx)	字符为红色，背景色为val		AA 02 00 62 00 00 0F 00 12 48
	99	63	VSP		FpgaDisplay(99, val, xx, xx)	字符为红色，背景色为val		AA 02 00 63 00 00 0F 00 12 48
	123	7B	IOVCC		FpgaDisplay(123, val, xx, xx)	字符为红色，背景色为val		AA 02 00 7B 00 00 0F 00 12 48
	124	7C	VCI		FpgaDisplay(124, val, xx, xx)	字符为红色，背景色为val		AA 02 00 7C 00 00 0F 00 12 48
组合类								
	30	1E	串格		FpgaDisplay(30, val, xx, xx)	中心为黑色(5/12h~7/12h,1/3v~2/3v). 1/4h~3/4h其它区域灰阶为val. 0~1/4h, 3/4h~h区域：黑白交替（4块）		AA 02 00 1E 00 7F 0F 00 12 48
	73	49	斜纹 （HTC SPEC）		FpgaDisplay(73, xx, xx, xx)			AA 02 00 49 0F 00 12 48
	79	4F	边框检测 （上下白左红右蓝）		FpgaDisplay(79, val, val1, val2)	中心为红色"WAKU"字符，背景色为val val1[6:4]定义上边线颜色，如100=red val1[2:0]定义下边线颜色,如010=green val2[6:4]定义左边线颜色,如001=blue val2[2:0]定义右边线颜色,如110=yellow		AA 02 00 4F 00 7F 00 77 00 41 0F 00 12 48

83	53	Crosstalk + Waku（waku线：左红右蓝上下白）		FpgaDisplay(83, val, xx, xx)	上边线白色，下边线白色，左边线红色，右边线蓝色。中心为黑色窗，背景色为val	AA 02 00 53 00 7F 0F 00 12 48
91	5B	Crosstalk + Waku（waku线：左红右蓝上下白）		FpgaDisplay(91, val, xx, xx)	上边线白色，下边线白色，左边线红色，右边线蓝色。中心为白色窗，背景色为val	AA 02 00 5B 00 7F 0F 00 12 48
100	64	crosstalk @ dot flicker, middle area		FpgaDisplay(100, val, val2, xx)	中心窗灰阶为val2. 其余部门以pixel为单位交替为：{val,0,val}和{0,val,0}	AA 02 00 64 00 7F 00 00 0F 00 12 48
101	65	crosstalk @ pixel flicker, middle area		FpgaDisplay(101, val, val2, xx)	中心窗灰阶为val2. 其余部门以pixel为单位交替为：{val,val,val}和{0,0,0}	AA 02 00 65 00 7F 00 00 0F 00 12 48
102	66	crosstalk @ column flicker, middle area		FpgaDisplay(102, val, val2, xx)	中心窗灰阶为val2. 其余部门以column为单位交替为：{val,0,val}和{0,val,0}	AA 02 00 66 00 7F 00 FF 0F 00 12 48
103	67	crosstalk @ pixel column flicker, middle area		FpgaDisplay(103, val, val2, xx)	中心窗灰阶为val2. 其余部门以column为单位交替为：{val,val,val}和{0,0,0}	AA 02 00 67 00 7F 00 FF 0F 00 12 48
136	88	snake1 hw spec		FpgaDisplay(136, xx, xx, xx)	华为snake1	AA 02 00 88 0F 00 12 48
137	89	snake2 hw spec		FpgaDisplay(137, xx, xx, xx)	华为snake2	AA 02 00 89 0F 00 12 48